

AMAS: Problem set 2



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Problem 1

1a

To produce best-fit parameters $(\hat{\mu}, \hat{\sigma}, \hat{\xi})$ from the data we use an unbinned likelihood approach using our own negative log likelihood function (LLH) and Minuit to minimize this.

To make life easier, we convert the HH:MM:SS format to seconds by simply splitting up the data. Upon inspecting the datafile, we see that the 10k time for Richard Suarez is offset by a column, so we fix this manually.

Fitting the GEV distribution on the data, **we obtain $\hat{\mu} = (2960 \pm 70)$, $\hat{\sigma} = (613 \pm 51)$, $\hat{\xi} = (-0.22 \pm 0.08)$, with errors from Minuit. We obtain a -LLH value of 1560.04.** The fit can be seen on fig. 1.

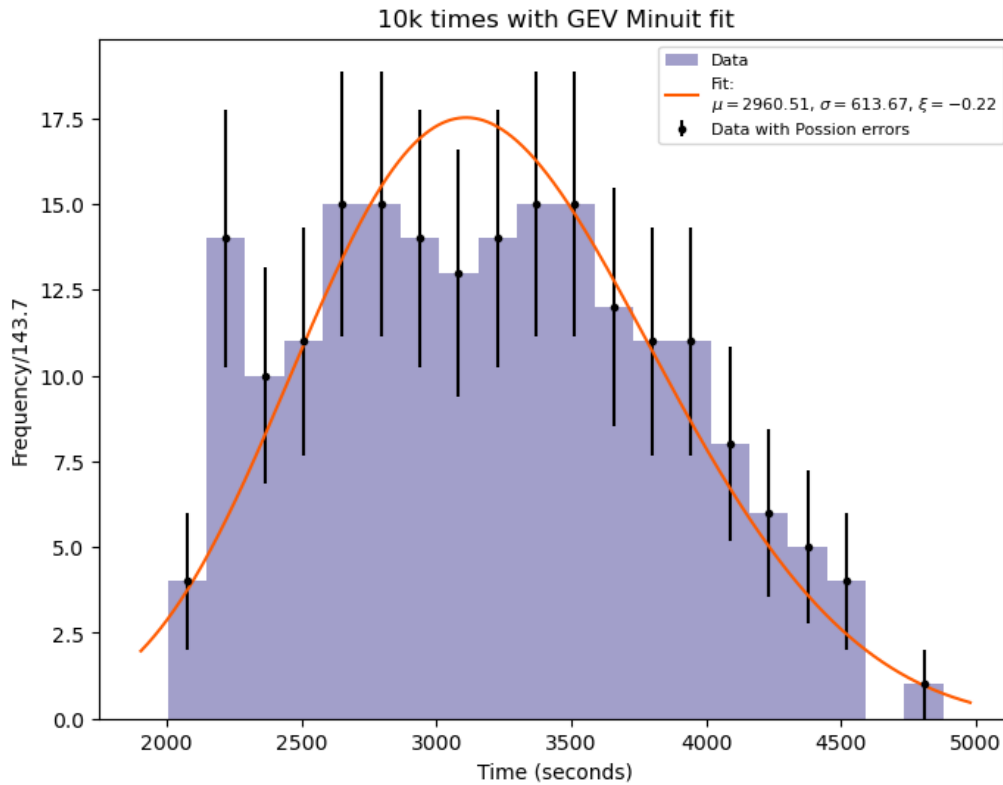


Figure 1: Minuit GEV fit on 10k running data.

But(!) Minuit assumes Gaussian errors - which is not necessarily the case. To make better error estimates, we use Bootstrapping using inverse MC sampling. The GEV distribution is a well-known distribution, with its CDF being:

$$y = F(x|\mu, \sigma, \xi) = \exp \left(- \left(1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right)^{-1/\xi} \right), \quad \xi \neq 0 \quad (1)$$

see [Wikipedia article about GEV](#).

Now to find $F^{-1}(x|\mu, \sigma, \xi)$, we solve for y in $x = F(y|\mu, \sigma, \xi)$:

$$x = \exp \left(- \left(1 + \xi \left(\frac{y - \mu}{\sigma} \right) \right)^{-1/\xi} \right) \quad (2)$$

$$\left(-\log(x)^{-\xi} - 1 \right) = \xi \left(\frac{y - \mu}{\sigma} \right) \quad (3)$$

$$y = \mu + \frac{\sigma}{\xi} \left(-\log(x)^{-\xi} - 1 \right) \quad (4)$$

Note: here y denotes the 10k times, and x is the CDF probability. In my code I use u to denote x to avoid confusion. Using the above equation, we can sample 10k times from the GEV distribution with our best fitted parameters, and refit to find the distributions of $\hat{\mu}, \hat{\sigma}, \hat{\xi}$, and the find the 1σ confidence intervals.

Sampling 1000 10k times and refitting, and repeating this process 2000 times, we get the following distributions on fig. 2.

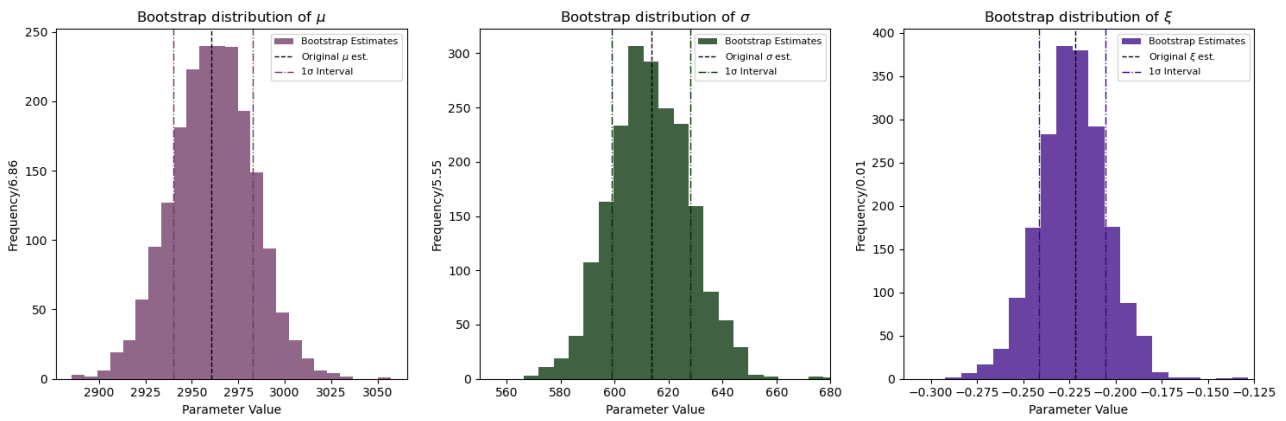


Figure 2: Bootstrapping distributions of $\hat{\mu}, \hat{\sigma}, \hat{\xi}$ based on 1000 samples simulated 2000 times.

We get slightly asymmetric errors on all the parameters. We can summarize our results in the table 1.

Parameter	LLH Estimate	Minuit error	Bootstrap error
$\hat{\mu}$	2960	± 70	(+22, -20)
$\hat{\sigma}$	613	± 51	(+14, -14.5)
$\hat{\xi}$	-0.22	± 0.08	(+0.01, -0.01)

Table 1: Table summarizing estimate for GEV parameters, and estimated errors via. Minuit and Bootstrapping approach.

The errors are slightly smaller via. Bootstrapping, but still on the same order of magnitude compared to Minuit.

1b

Retriving the 10k time of SomethingAboutGoat (SAG), we can find the probability of running slower than SAG as $1 - P(x < x_{SAG})$, i.e. 1 minus the probability of being faster than SAG. We simply use SciPys quad function and our best estimates for the GEV parameters, to integrate $f(x|\hat{\mu}, \hat{\sigma}, \hat{\xi})$ from 0 to x_{SAG} .

We get $P(x > x_{SAG})_{EST} = 0.1985 \approx 19.85\%$.

Using the same approach, but here just summing, since we have discrete data we can find the same probability based on the data.

We get $P(x > x_{SAG})_{DATA} = 0.2323 \approx 23.23\%$.

Upon looking at the two figures, they seem okay close. The percent-wise difference is:

$$\frac{0.1985 - 0.2323}{0.2323} = -0.1453 \quad (5)$$

i.e. using our best fit parameters, we get a probability which is about 14.5% smaller than what our data indicates. They surely disagree within any reasonable bound. Looking at table 1 it might be possible within our errors to find estimates for the fit parameters where the probabilities agree more.

1c

Simply locking the parameters, and again using quad on the GEV function using the said parameters, we can obtain the CDF.

The plot of the CDF with the histogrammed data can be seen on fig. 3.

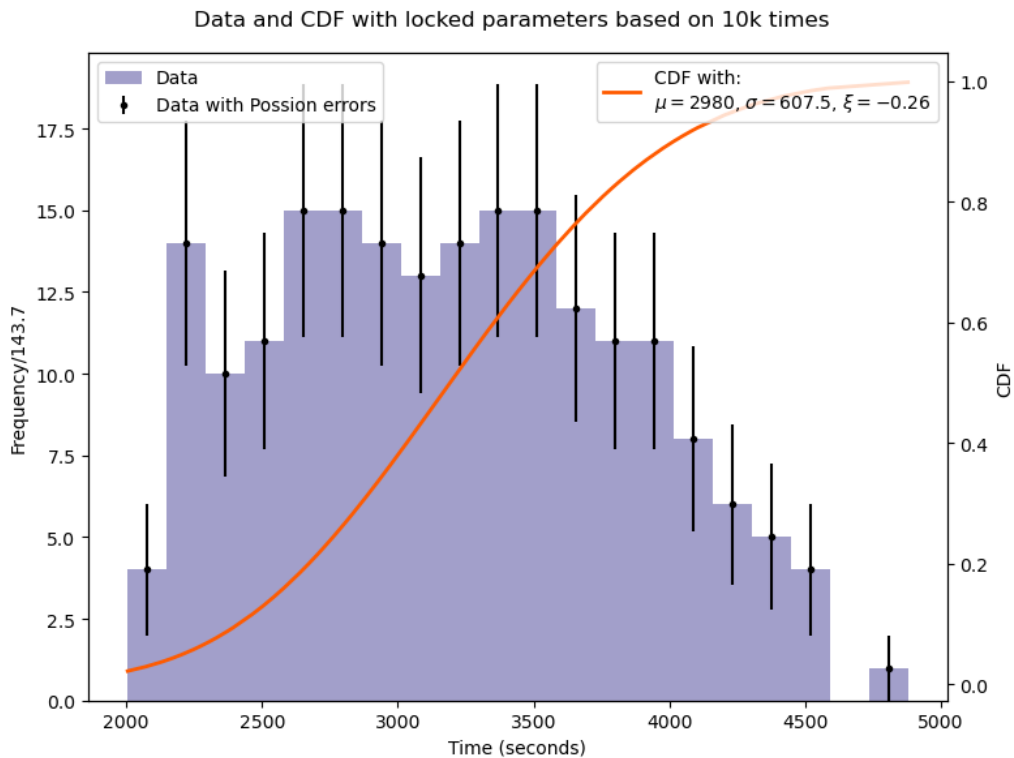


Figure 3: Figure showing the data and CDF based on locked parameters and 10k times.

The probability of getting a quicker time than the fastest time in the 10k dataset, can again be found using quad.

We find the probability of getting a faster time than the fastest time with the locked parameters, to be $P(x < x_{min}) = 0.022 \approx 2.2\%$.

Problem 2

Let us start by normalizing the prior function for $x \in (0, 4)$:

$$1 = \int_0^4 dx f(x) = \int_0^{1.6} dx 1/N + \int_{1.6}^{2.1} dx 2.66/N + 0 \quad (6)$$

$$= 1.6/N + 2.66(0.5)/N = 1.6/N + 1.33/N \quad (7)$$

$$= 2.93/N \quad (8)$$

$$\Rightarrow N = 2.93 \quad (9)$$

i.e. $N=2.93$ for $f(x)$ to be normalized.

The likelihood, i.e. the Γ -function, can easily be normalized using quad again, so we just define the non-normalized Γ -function and defining a new, which normalizes it.

We then calculate the prior and likelihood values for $x \in (0, 4)$, and then the posterior as the product. **This posterior is once again normalized using quad.**

Using XKCD colors (like above), we get the plots on fig. 4.

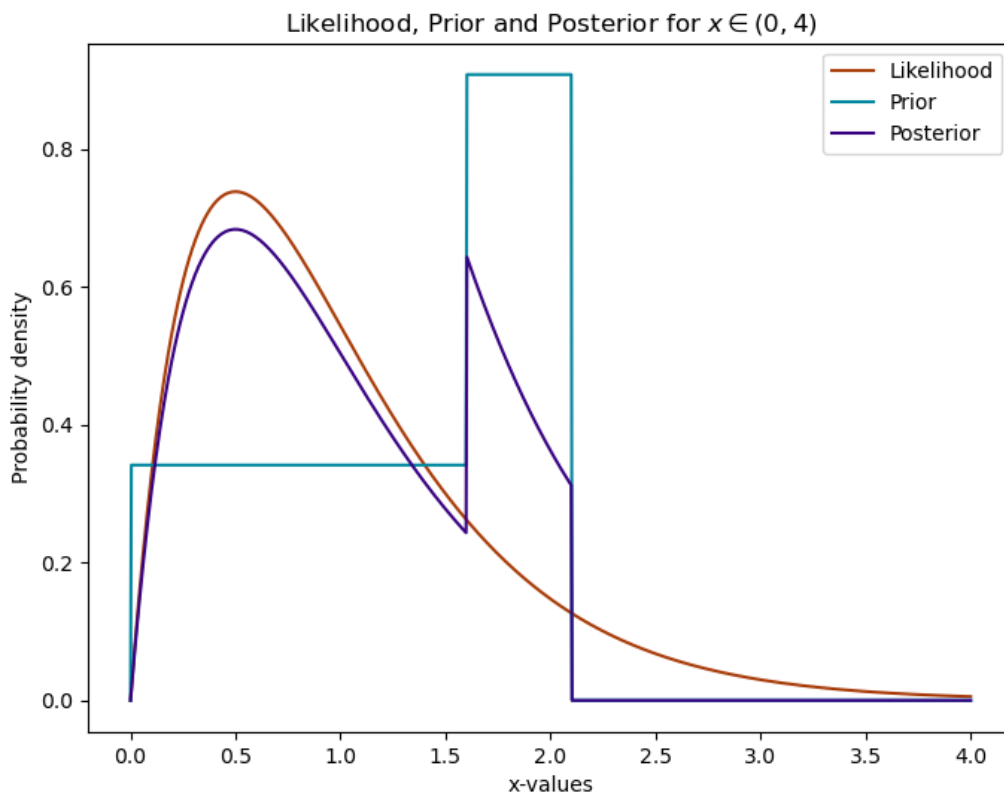


Figure 4: Plots of likelihood, prior and posterior based on custom prior and gamma likelihood for $\alpha = \beta = 2$.

Based on the posterior, we find $\hat{x} = 0.501$ using argmax , and we get $g(\hat{x}) = 0.6833$.

Problem 3

3a

This problem is solved by a for-loop, where for each day, we choose a random angle $\theta \in [0, 2\pi]$ and move $r_{max} = 0.2$ km in that direction - as a first proposal. To keep track of the crab and ensure that it doesn't wander off the island, we calculate the distance from origo to the new point. If the new point is situated outside the island, we calculate how far there is to the border via parametrization:

$$|P_0 + tP_1|^2 = r_{island}^2 \quad (10)$$

where P_0 is the former position, P_1 is the proposed new position and t is the parametrization parameter ($t \in [0, 1]$, i.e. we find which t is needed to hit the border).

It is a quadratic equation in t :

$$|P_0|^2 - r_{island}^2 + t^2|P_1|^2 + 2tP_0P_1 = 0 \quad (11)$$

so this is solved, and the new position is calculated and stored.

For a crab starting at (3.6, -2.0) km, we get the following path for 200 days shown on fig. 5.

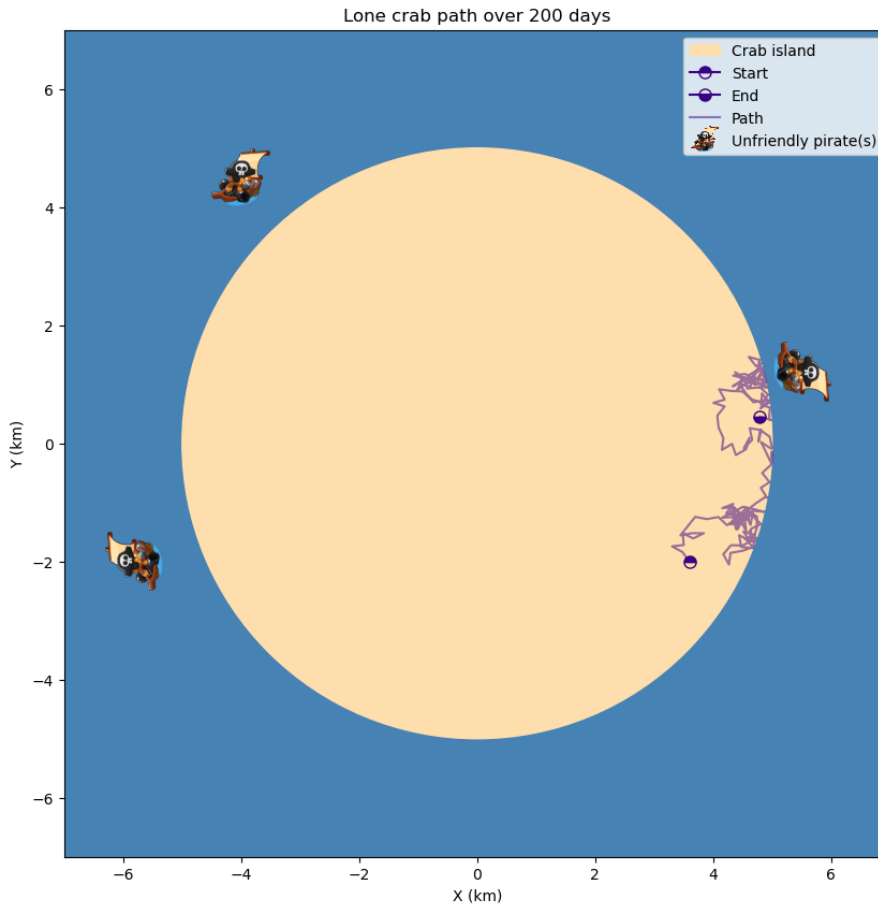


Figure 5: Path of a lone crab on the island after 200 days.

3b

We reuse our function from 3a and modify it, such that it keeps running until the crab has hit the barrier once. The total distance travelled is then the sum of distances between each point, which is calculated using `linalg norm`. We do this 501 times.

We then get the distribution shown on fig. 6.

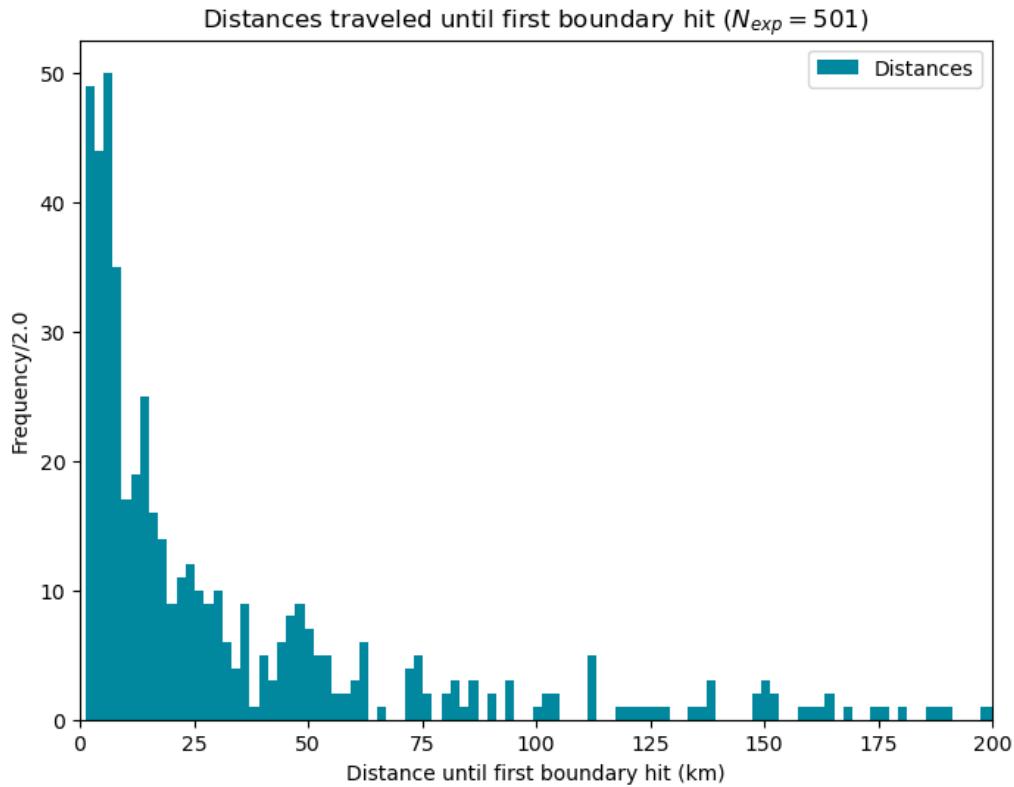


Figure 6: Distances until the crab hits the edge of the island.

3c

We can reuse our single crab walk function for this problem to evolve the position of each crab each day. To keep track of the alive crabs and their masses we create two arrays, one boolean (alive = True, dead = False) and one full of ones to modify.

Then for each day, we create so-called battle pairs, i.e. the pairs of crabs that are in battle range (inter distance $< 0.175\text{km}$), and save the array. We can then sort the array in terms of smallest to largest distance. Then for each battle pair, we calculate the winning odds of the largest crab, and choose a random number, $x \in [0, 1]$, and if $x \leq P_{large}$, the largest crab wins and consumes the smallest - here we change the status of the small crab in our alive-array from True to False. If $x \not\leq P_{large}$, we don't change anything and both crabs remain at that position until the next day.

To verify that our function works, we have visualized the 200 day crab battle on fig. 7. The cross signifies a dead crab.

To determine the most likely mass of the largest crab and number of individual crabs alive after 200 days, we simulate 10000 crab battles and find the mode.

The results are visualized on fig. 8

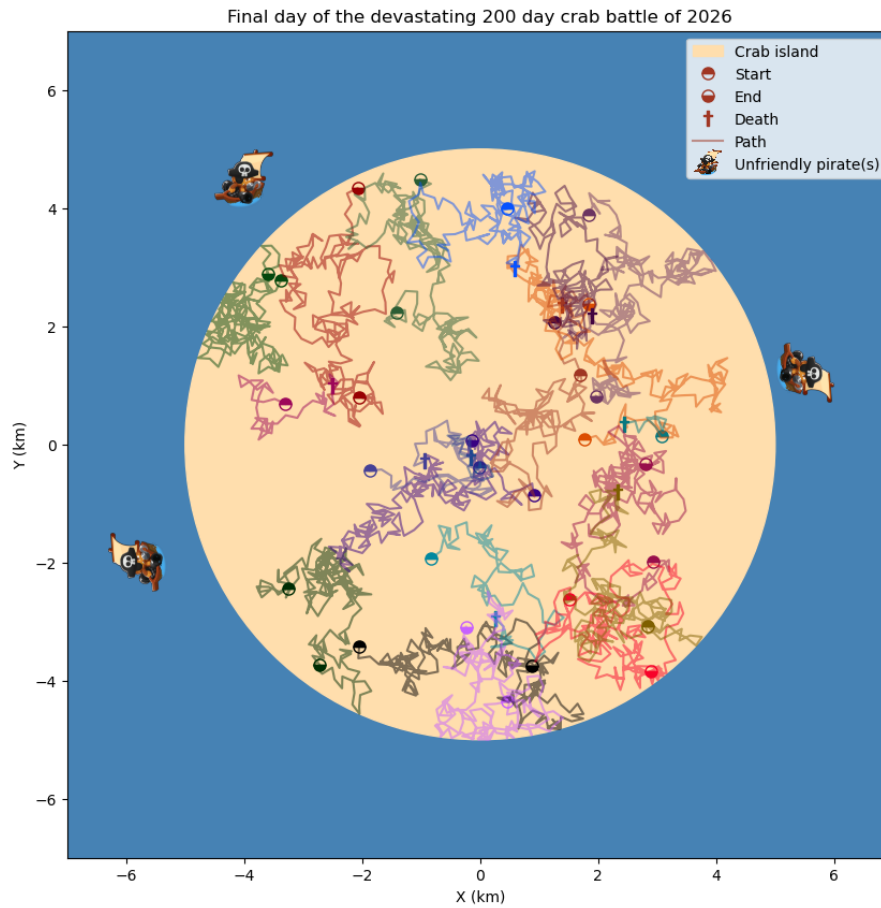


Figure 7: Scatter plot of the 200 day crab battle with the starting positions provided and battle logic by the exercise description.

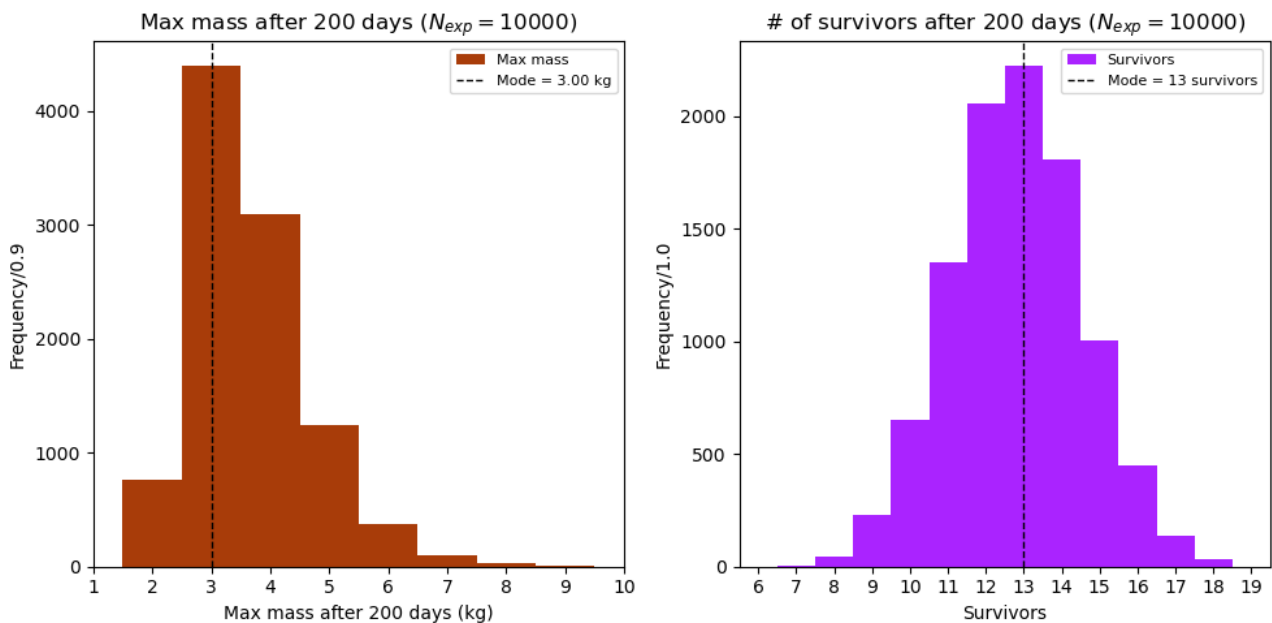


Figure 8: Distributions of mass of largest crab alive and number of surviving crabs after 200 days.

As can be seen, the most likely number of surviving crabs after 200 days is 13, and the most likely mass of the largest crab is 3.00 kg.

3d

The solution to this problem is inspired by 3b, i.e. we simply modify our solution to 3c by enforcing a while-loop that runs until the number of alive crabs is 10 and save the amount of days, and do this 10000 times. The distribution is highly skewed, so we manually calculate the 1σ confidence interval, by finding the lower and upper tail containing 15.85% and 84.15% of the distribution, respectively.

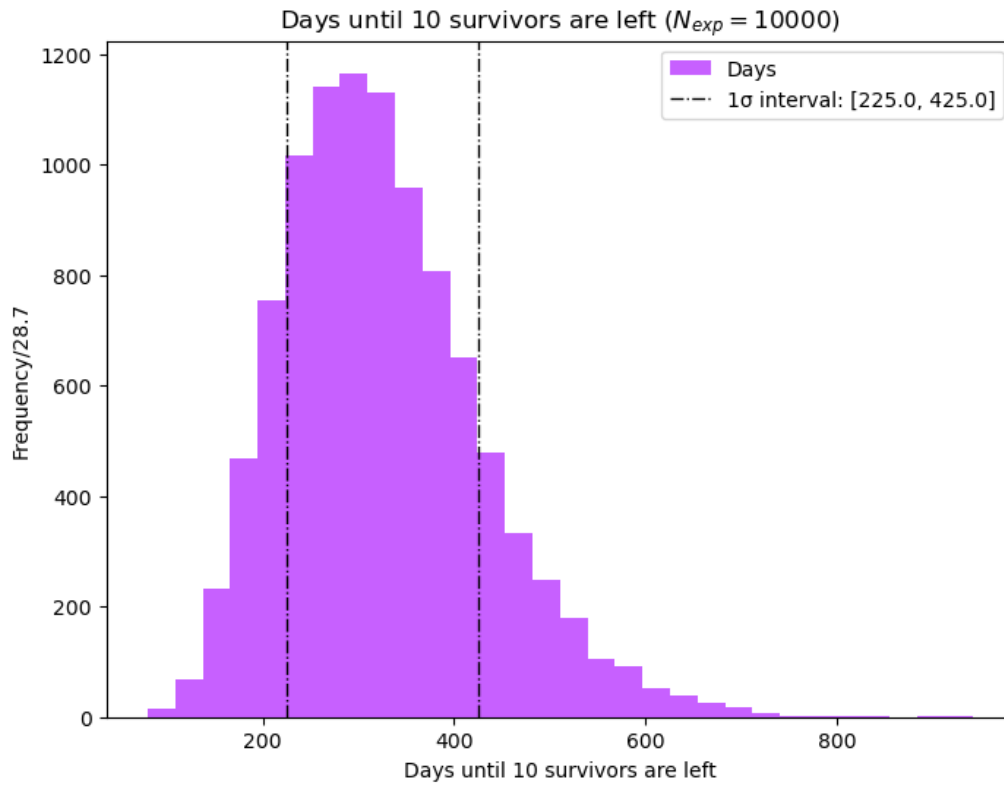


Figure 9: Distribution of the amount of days until there a only 10 crabs alive based on 10000 experiments.

As can be seen, **the 1σ confidence interval is [225.0, 425.0].**